



ONYX

MEASURING ARM -
PORTABLE CMM -
ONYX

MORE ERGONOMIC
LIGHTER
MORE FLEXIBLE
MORE ACCURATE
MORE RELIABLE



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BATTERY CHARGE

Battery life:
8 hours probing
4 hours scanning.
Hot swappable battery.

AMBIENT TEMPERATURE

Checking and monitoring of
ambient temperature:
Prevent use outside normal
operating temperatures (10-45°C).
Compensate for the expansion of
parts during significant
temperature variations.





ARM STATUS



WIFI AND BLUETOOTH

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SECURE REST POSITION

A flexible fin cushions the return of the arm to its rest position. A magnet then secures the arm in its rest position and prevents it from swinging forwards.



120° RANGE ON AXIS 2

The large range of axis 2 enables the arm to reach very low areas. A particularly useful feature for tools inspections.



HANDLE LOCK

To prevent the handle moving along the arm, it is fixed by magnet to axis 2. This lock prevents the scanner from hitting the arm, for example.

EASYCLIP



INCREASED FLEXIBILITY

The handle of the Onyx 7-axis arm can be removed to enable measurement in cavities and areas that would otherwise be inaccessible. The handle is attached and removed simply by pressing the unlock button.



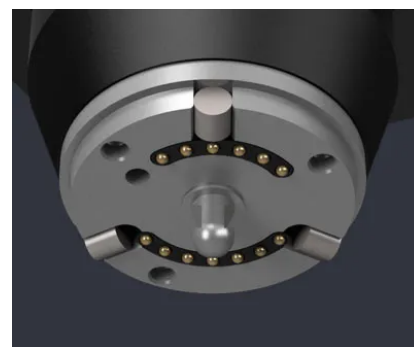
INTUITIVE CONTROL

Scanning and probing operations are controlled intuitively using highly responsive buttons located in ideal positions on the handle.



SECURE HOLD

The handle grip offers a secure hold, essential for preventing slipping during prolonged use of the arm.



REPEATABILITY OF THE RENISHAW INTERFACE

The probe can be changed without recalibration thanks to the repeatability of the Renishaw interface. The arm automatically detects the probe being used.



PROBE OR SCAN, WHY CHOOSE?

With the Onyx arm, you can choose the best acquisition method, either probing or scanning, depending on the type of part, quickly, at any time and without recalibration.

MPLEMENTARITY
OF MEASURES



learned with the 3D scanner without
interrupting the workflow.

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THE POWER OF SKYLINE SCANNERS

Perfectly integrated into the Onyx arm,
Skyline 3D scanners meet the most
diverse 3D scanning requirements.

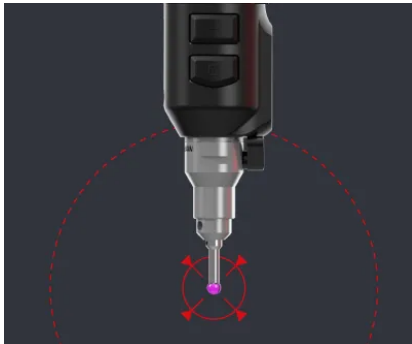


INCREASED PRODUCTIVITY THANKS TO

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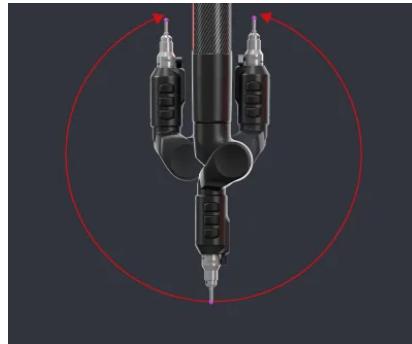
SCANNER

For even greater flexibility, there's no need to remove the scanner to start probing operations, as a probe can be positioned directly under the Kreon scanners.



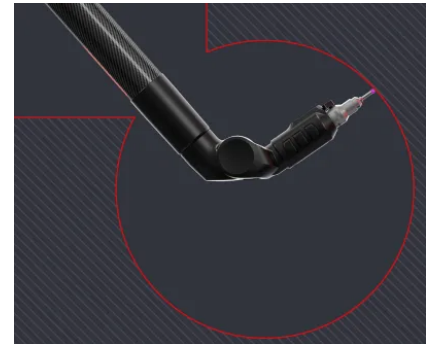
EXCEPTIONAL ACCURACY

The 6-axis version of the Onyx arm is up to 20% more accurate than the 7-axis version. This version of the arm is the most accurate in the Kreon range.



FREEDOM OF MOVEMENT

Axis 6 is extremely mobile with an impressive 360° range. It can easily reach the underside of certain assemblies or the inside of rabbets.



PROBING DIFFICULT- TO-ACCESS AREAS

The interior of certain parts, impossible to measure with tracked tools, can easily be reached with the fineness of the arm's ending.



HIGH- RESOLUTION ENCODERS

To offer an exceptional level of accuracy, high-resolution encoders are integrated in the Onyx measuring arm. In this way, the angle formed by each axis of the arm is better determined and each position of the arm during movement is calculated with great precision.



BACK PANEL

1 BATTERY ACCESS COVER

Hot swappable battery:
can be removed without
switching off the
measuring arm and
without interrupting the
workflow.

2 BATTERY VENTILATION GRID

To preserve battery life.

3 POWER SUPPLY FOR THE ARM

International standard
100-250v power supply.

4 ETHERNET CONNECTION

When WiFi and Bluetooth
are disabled, or when
these protocols cannot be
used for security reasons,
data from the scanner and
the measuring arm can be
transferred by Ethernet.



ULTRA LIGHT CARBON STRUCTURE

Onyx arm is a high-tech product made from materials used in the aeronautics industry. Carbon fibre is used intensively, ensuring stability while remaining extraordinarily light.



OPTIMAL ACCURACY, WHATEVER THE CONDITIONS



TEMPERATURE SENSORS

Sensors in the arm ensure that the temperature is constantly monitored and compensated, so that any expansion of the arm components, associated with temperature variations, does not affect measurement accuracy.



COMPLIES WITH THE ISO 10360-12 STANDARD

All Onyx measuring arms are tested in accordance with the ISO 10360-12 standard. This standard guarantees that the measurement accuracies specified are respected during testing.

What is the ISO standard ?



ENDPOINTS IN THE SOFTWARE FOR AXES 2, 4 AND 6

When axes 2, 4 and 6 are constrained, they can cause measurement deviations. To avoid this, measurements are automatically stopped when they approach the endpoints of these axes. However, these limitations can be disabled if required.

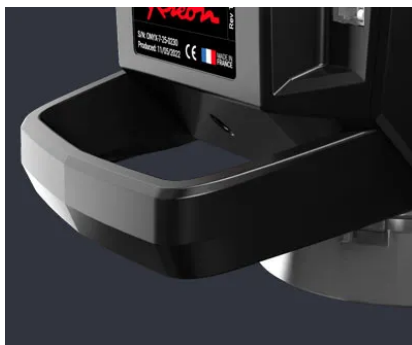


SPEED OF INSTALLATION



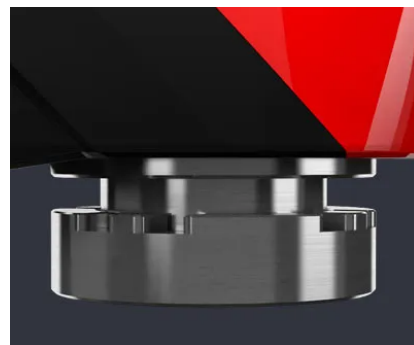
COMPACT AND ROBUST TRANSPORT CHEST

The chest on wheels supplied with the portable measuring arm protects the arm during transport. It also makes it easy to move the arm as close as possible to the application.



BACK HANDLE

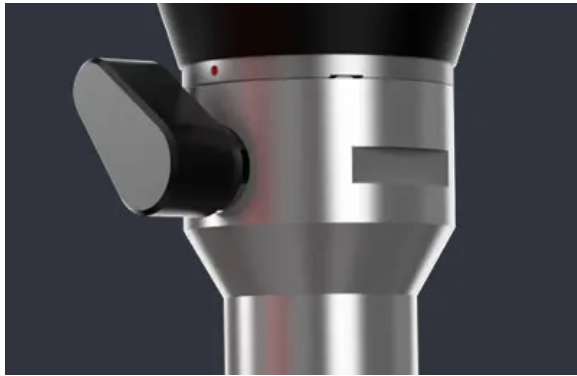
The handle on the back of the arm provides a secure grip for easy removal from the chest and placement on the arm support.



BRUNSON INTERFACE

The Brunson universal interface can be quickly fastened to various supports such as tripods, magnetic bases or rolling carts, to provide a secure hold.

DESIGNED FOR USE WITHOUT TOOLS



PROBES WITH INTEGRATED KEY

The probes supplied with the arm are equipped with an integrated key for faster assembly.



MAGNETISED BATTERY COVER

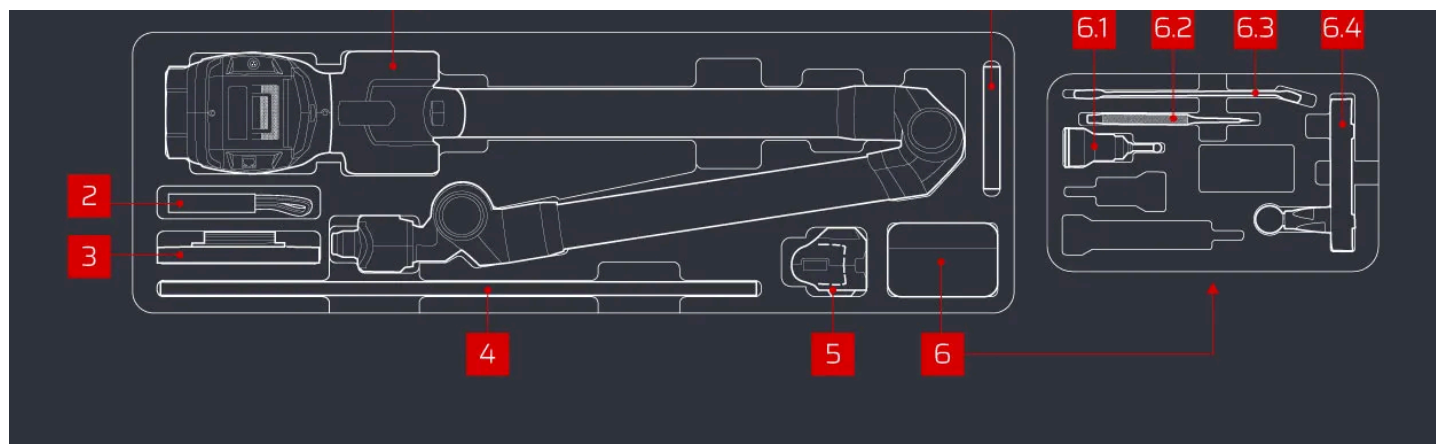
The battery cover is simply magnetised, allowing easy access to the battery for replacement or recharging.



USE OF LEAPFROG FOR LARGE PARTS

The leapfrog technique is used to move the arm a little at a time in order to measure large parts whose dimensions exceed the range of the measuring arm. To move from one measuring position to another, the operator simply needs to measure three references common to both positions. The Kreon **leapfrog kits** are available on request.

CONTENTS OF THE TRANSPORT SUITCASE



- ▶ Onyx measuring arm
- ▶ Ethernet cable, power and dust cover
- ▶ Base
- ▶ Skyline scanner (option)
- ▶ Accessory kit (1. probe 6 mm, 2. stylus tool, 3. flat key, 4. calibration sphere for probe)
- ▶ Kreon box (Zenith software, Kreon dongle and calibration certificates)

YOU ARE INTERESTED IN THE ONYX MEASURING ARM

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OTHER REQUEST





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TECHNICAL SPECIFICATIONS

ONYX 6-AXIS



ONYX 7-AXIS



Arm model	E Uni	P Size	P Form	L Dia	SPAT
Onyx-7-20	0.024 mm	0.010 mm	0.018 mm	0.040 mm	0.018 mm
Onyx-7-25	0.027 mm	0.011 mm	0.021 mm	0.044 mm	0.020 mm
Onyx-7-30	0.043 mm	0.016 mm	0.028 mm	0.072 mm	0.032 mm
Onyx-7-35	0.053 mm	0.020 mm	0.035 mm	0.087 mm	0.043 mm
Onyx-7-40	0.068 mm	0.025 mm	0.040 mm	0.102 mm	0.052 mm
Onyx-7-45	0.100 mm	0.038 mm	0.049 mm	0.110 mm	0.065 mm
Onyx-7-50	0.120 mm	0.052 mm	0.062 mm	0.125 mm	0.089 mm

MAXIMAL REACHABLE DISTANCE AND MEASURING VOLUME





Arm model*	Measuring volume (diameter)	Maximum reachable distance (diameter)
Onyx-20	2 m	2.42 m
Onyx-25	2.5 m	2.92 m
Onyx-30	3 m	3.42 m
Onyx-35	3.5 m	3.92 m
Onyx-40	4 m	4.42 m
Onyx-45	4.5 m	4.92 m
onyx-50	5 m	5.42 m

*Same data for both 6 and 7-axis versions.



SOLUTIONS

Measuring arms

Onyx
Onyx Skyline
Ace
Ace Skyline
Baces

Scanning solutions

Scanning CMMs
Track Ace

3D scanners

Skyline
Zephyr III

Software

Zenith
Kreon Plugins

APPLICATIONS

Quality control
Design and R&D
Reverse engineering
CAD comparison
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Maintenance and repair
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